

Groups, rings and modules

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Lecture 1 – 22/01/26

§1 Groups

§1.1 Groups – a reminder

Definition 1.1. A *group* is a triple $(G, \cdot, e)$ where G is a set,

$$\cdot : G \times G \rightarrow G$$

is a function, and $e \in G$ such that

- *Associativity.* $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
- *Identity.* For all $a \in G$, $a \cdot e = e \cdot a = a$;
- *Inverses.* For all $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

The size of G , $|G|$, is called the *order* of the group.

Proposition 1.2 (Uniqueness of inverses)

Inverses are unique.

Proof. Left as an exercise (see 1A). ■

Definition 1.3 (Subgroups). If G is a group and $H \subseteq G$, then we say H is a *subgroup* if

- $e \in H$;
- for all $a, b \in H$, $a \cdot b \in H$;
- $(H, \cdot, e)$ is a group.

In order to say item 3, we must first establish 1 and 2.

Proposition 1.4

A non-empty subset $H \subseteq G$ is a subgroup if and only if, for all $h_1, h_2 \in H$, $h_1 h_2^{-1} \in H$.

Proof. Left as an exercise (see 1A). ■

Definition 1.5. A group G is called *abelian* if, for all $g_1, g_2 \in G$, $g_1 g_2 = g_2 g_1$.

Example 1.6 • We have $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}/n\mathbb{Z}$.

- The symmetric groups S_n : permutations of $\{1, \dots, n\}$ under composition. More generally for any set X , we have $\text{Sym}(X)$.
- The dihedral groups D_{2n} : the group of symmetries of a regular n -gon.
- The general linear group over $\mathbb{R}$, $\text{GL}(n, \mathbb{R})$ or $\text{GL}_n(\mathbb{R})$, is the set of $\{f : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid f \text{ is an } \mathbb{R}\text{-linear bijection}\}$. Since $\text{GL}(n, \mathbb{R})$ is made of functions, we can think of it a subset of $\text{Sym}(\mathbb{R}^n)$. (Equivalently, it's the set of n by n invertible matrices with elements in $\mathbb{R}$).

Some examples of subgroups:

- The even permutations $A_n \leq S_n$ (the alternating group);
- Rotational symmetries of an n -gon, $C_n \leq D_{2n}$;
- Determinant one matrices $\text{SL}(n, \mathbb{R}) \leq \text{GL}(n, \mathbb{R})$.

Other examples include $C_2 \times C_2$, $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Definition 1.7. Let G be a group, $g \in G$ and $H \leq G$ a subgroup. Then the *left coset*

$$gH = \{gh : h \in H\}$$

Basic properties include that $g \in gH$, so cosets are non-empty. Also, cosets partition G – they have no non-trivial overlaps and every g lies in some coset. Furthermore, the map

$$\begin{aligned} H &\rightarrow gH \\ h &\mapsto gh \end{aligned}$$

defines a bijection.

Theorem 1.8 (Lagrange)

If G is a finite group and $H \leq G$ then

$$|G| = |H| \cdot |G : H|$$

where $|G : H|$, the index of H in G , is the number of cosets of H in $|G|$.

Definition 1.9. Let G be a group and let $g \in G$. The *order* of g , $\text{ord}(g)$, is the smallest positive integer n such that $g^n = e$, or infinity if no such n exists.

Proposition 1.10

Let $g \in G$. Then, $\text{ord}(g) \mid |G|$.

Proof. Let $g \in G$ and consider $H = \{e, g, g^2, \dots, g^{n-1}\}$ where $n = \text{ord}(g)$. This is a subgroup: it is non-empty, and $g^r \cdot g^{-s} = g^{r-s} \in H$ (after potentially multiplying by $g^n = e$). Also, the elements of H are distinct – if $g^i = g^j$ for $0 \leq i < j \leq n$, then $g^{j-i} = e$, contradicting the minimality of $\text{ord}(g)$. Hence $|H| = \text{ord}(g)$ – now apply Lagrange. ■

§1.2 Normal subgroups, quotients and homomorphisms

Let $H \leq G$. When are two cosets of H in G equal? If $gH = g'H$, then $g \in g'H$, so $g = g'h$ for some $h \in H$. In other words, $gH = g'H \iff (g')^{-1}g \in H$.

We denote the set of left cosets of $H \leq G$ by $G/H = \{gH : g \in G\}$. Does G/H carry a natural group structure?

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We propose the ‘natural’ group structure:

$$g_1H \cdot g_2H = (g_1g_2)H$$

But this rule may depend on the coset representatives we pick. We must check that it doesn't. Suppose $g_2H = g'_2H$, so $g'_2 = g_2h$ for some $h \in H$. Then

$$g_1H \cdot g'_2H = (g_1g'_2)H = g_1g_2hH = g_1g_2H$$

On the other hand, if $g_1H = g'_1H$, so $g'_1 = g_1h$ for $h \in H$. Then

$$g'_1H \cdot g_2H = g_1hg_2H$$

and that is not automatically equal to g_1g_2H – we need $g_2^{-1}hg_2 \in H$ to hold for all $g_2 \in G$, $h \in H$.

Definition 1.11 (Normal subgroups). A subgroup $H \leq G$ is *normal* if, for all $g \in G$, $h \in H$, $ghg^{-1} \in H$. We write this as $H \triangleleft G$.

Remark 1.12. When $H \triangleleft G$, then G/H inherits a group structure by the rule above. We will refer to this as the *quotient group of G by H* .

Definition 1.13 (Homomorphisms). Let G and H be groups. A homomorphism from G to H is a function $f : G \rightarrow H$ such that, for all $g_1, g_2 \in G$, $f(g_1g_2) = f(g_1)f(g_2)$.

Results such as $f(e_G) = e_H$ follow immediately.

Lemma 1.14

If $f : G \rightarrow H$ is a homomorphism, then $f(g^{-1}) = f(g)^{-1}$.

Proof. Calculate $f(g \cdot g^{-1})$ in two ways.

$$\textcircled{1} \quad f(gg^{-1}) = f(e_G) = e_H;$$

$$\textcircled{2} \quad f(gg^{-1}) = f(g)f(g^{-1})$$

so $f(g)f(g^{-1}) = e_H$ and the result follows. ■

Definition 1.15. Let $f : G \rightarrow H$ be a homomorphism. The *kernel* of f is

$$\ker f = \{g \in G : f(g) = e_H\}$$

and the *image* of f is

$$\text{im } f = \{h \in H : h = f(g) \text{ for some } g\}$$

Proposition 1.16

Let $f : G \rightarrow H$ be a homomorphism. Then $\ker f \triangleleft G$ and $\text{im } f \leq H$.

Proof. Observe $e \in \ker f$ so $\ker f$ non-empty. Now, if $x, y \in \ker f$, then

$$f(xy^{-1}) = f(x)f(y)^{-1} = e$$

so $xy^{-1} \in \ker f$. For normality, let $x \in G$ and $g \in \ker f$. Now

$$f(xgx^{-1}) = f(x)ef(x)^{-1} = e$$

so $xgx^{-1} \in \ker f$ and hence $\ker f \triangleleft G$.

For the image, observe that, if $a, b \in \text{im } f$ with $a = f(x)$, $b = f(y)$, then $ab^{-1} = f(xy^{-1}) \in \text{im } f$. ■

Remark. Observe $f : G \rightarrow G/H$ taking $g \mapsto gH$ is a homomorphism. This is a nice way of expressing its ‘naturalness’.

Definition 1.17. A homomorphism $f : G \rightarrow H$ is an *isomorphism* if it is a bijection. Two groups G and H are said to be *isomorphic*, $G \cong H$, if there exists an isomorphism between them.

Theorem 1.18 (First isomorphism theorem)

Let $f : G \rightarrow H$ be a homomorphism, then $\ker f$ is normal in G and the homomorphism

$$\begin{aligned} G/\ker f &\rightarrow \operatorname{im} f \\ g\ker f &\mapsto f(g) \end{aligned}$$

is an isomorphism. In particular, $G/\ker f \cong \operatorname{im} f$.

Proof. We have already shown the normality result. Consider the map

$$\begin{aligned} \varphi : G/\ker f &\rightarrow \operatorname{im} f \\ g\ker f &\mapsto f(g) \end{aligned}$$

- φ is well-defined. If $g\ker f = g'\ker f$, then $g'g^{-1} \in \ker f$, so $f(g'g^{-1}) = e$ which gives $f(g) = f(g')$ as required.
- φ is surjective. Let $h \in \operatorname{im} f$. Then $h = f(g)$ for some $g \in G$, and so $h = \varphi(g\ker f)$.
- φ is injective. Suppose $\varphi(g\ker f) = \varphi(g'\ker f)$, then $f(g) = f(g')$. But then $g'g^{-1} \in \ker f$, so $g\ker f = g'\ker f$.

■

Example

Consider the homomorphism

$$\begin{aligned} \exp : (\mathbb{C}, +) &\rightarrow (\mathbb{C}^*, \times) \\ z &\mapsto e^z \end{aligned}$$

Then $\ker \exp = 2\pi i\mathbb{Z}$, so $\mathbb{C}^* \cong \mathbb{C}/2\pi i\mathbb{Z}$.

Theorem 1.19 (Second isomorphism theorem)

Let $H \leq K$ and $K \triangleleft G$. Then

$$HK = \{hk : h \in H, k \in K\}$$

is a subgroup of G . Furthermore, $H \cap K$ is normal in H , and there is an isomorphism $HK/K \cong H/H \cap K$.

Proof. We will show the results in order.

- HK is a subgroup. It contains e_G , so is non-empty. Now consider $x = hk$ and $y = h'k'$. We'll show $yx^{-1} \in HK$. Observe

$$\begin{aligned} yx^{-1} &= h'k'(hk)^{-1} \\ &= h'k'k^{-1}h^{-1} \\ &= (h'h^{-1})h(k'k^{-1})h^{-1} \end{aligned}$$

$$= h'h^{-1}k''$$

by normality of K .

- $H \cap K$ is a kernel. Consider

$$\begin{aligned}\varphi : H &\rightarrow G/K \\ h &\mapsto hK\end{aligned}$$

by restricting the domain from $G \rightarrow G/K$ – the kernel is $H \cap K$.

- *Final statement.* By the first isomorphism theorem, we have

$$\text{im } \varphi \cong H/(H \cap K)$$

so we must show $\text{im } \varphi \cong HK/K$. But $\text{im } \varphi$ is the set of K -cosets in G that are representable by elements of H , i.e. $\{hK : h \in H\} \subseteq G/K = HK/K$. ■

§1.2.1 Correspondence between subgroups of G and G/K

Let $p : G \rightarrow G/K$ be the quotient homomorphism. We have a bijection

$$\begin{aligned}\{\text{subgroups of } G \text{ containing } K\} &\leftrightarrow \{\text{subgroups of } G/K\} \\ K \triangleleft L \leq G &\mapsto L/K \leq G/K \\ \{g \in G : gK \in A\} &\leftrightarrow A \leq G/K\end{aligned}$$

The forward bijection is just applying the projection map on a subset, and, since the quotient map is a homomorphism, its image is a subgroup; and the backward bijection is actually just applying the preimage of p . It is an exercise to check that the forward and backward directions are inverses.

Remark. In fact the correspondence matches normal subgroups on the two sides.

Theorem 1.20 (Third isomorphism theorem)

Let $K \leq L \leq G$ be normal subgroups of G . Then there exists an isomorphism $(G/K)/(L/K) \cong G/L$.

Proof. Define a map

$$\begin{aligned}\varphi : G/K &\rightarrow G/L \\ gK &\mapsto gL\end{aligned}$$

- φ is well-defined. If $gK = g'K$, then $g'g^{-1} \in K \subseteq L \implies gL = g'L$.
- φ is a homomorphism. Clear.

We have $\ker \varphi = \{gK : gL = L\} = \{gK : g \in L\} = L/K$. Hence we are done by the first isomorphism theorem. ■

Definition 1.21 (Simplicity). A group G is called *simple* if its only normal subgroups are G and $\{e\}$.

Proposition 1.22

Let G be an abelian group. Then G is simple if and only if $G \cong C_p$ for some prime p .

Proof. If $G \cong C_p$, then any non-identity element generates G by, for example, Lagrange. So G is simple.

If G is simple and abelian, then take any $g \neq e \in G$ and consider $\langle g \rangle$. This is a subgroup, and since G is abelian, it is automatically normal. Hence, by simplicity, $\langle g \rangle = G$, and so G is cyclic.

If G is infinite and cyclic, then $G \cong \mathbb{Z}$, but $\mathbb{Z}$ is not simple, so this is a contradiction. Hence G is finite and $G \cong C_m$. If q divides m , take the subgroup generated by $g^{m/q}$, where g generates G . Normality is automatic, so $q = 1$ or $q = m$ by simplicity – in other words, m is prime. ■

Theorem 1.23

Let G be a finite group. Then, there exist subgroups

$$G = H_1 \triangleright H_2 \triangleright H_3 \triangleright \cdots \triangleright H_n = \{e\}$$

such that H_i/H_{i+1} is simple for all i .

Proof. If G is simple, $H_2 = \{e\}$, and we're done. Otherwise, let H_2 be a proper normal subgroup of maximal order in G . We claim G/H_2 is simple. If not, consider the quotient map $\pi : G \rightarrow G/H_2$.

If $N \triangleleft G/H_2$ is a proper normal subgroup, by the correspondence theorem, we can find a proper normal subgroup in G containing H_2 . But this contradicts the maximality of H_2 .

Now we can just iterate – note that $|H_{i+1}| < |H_i|$, so we will eventually terminate. ■

Remark. This sequence is not necessarily unique.

§1.3 Group actions and permutations

Definition 1.24. Let X be a set. Let $\text{Sym}(X)$ be the set of permutations on X . The identity is the identity function, and the group law is given by composition. If $X = \{1, \dots, n\}$, then we write $\text{Sym}(X) = S_n$.

- It's convenient to write $\sigma \in S_n$ as a product of disjoint cycles.
- Every $\sigma \in S_n$ is a product of transpositions, and the number of such transpositions gives rise to the *sign* of σ .

$$\text{sign} : S_n \rightarrow \{\pm 1\}$$

$$\sigma \mapsto \begin{cases} +1 & \sigma \text{ is a product of an even number of transpositions} \\ -1 & \sigma \dots \text{ odd} \dots \end{cases}$$

sign is a homomorphism, and is surjective when $n \geq 3$.

Let X be a set and G a group, and consider a homomorphism $G \rightarrow \text{Sym}(X)$. This is called a *permutation representation* of G .

Definition 1.25 (Group actions). An *action* of a group G on a set X is a function

$$\begin{aligned} G \times X &\rightarrow X \\ (g, x) &\mapsto g \cdot x \end{aligned}$$

such that

- $e \cdot x = x$ for all $x \in X$;
- $g_1 \cdot (g_2 x) = (g_1 g_2) \cdot x$ for all $g_1, g_2 \in G$ and $x \in X$.

We say $G \curvearrowright X$.

We should note the bijection between permutation representations and group actions.

Proposition 1.26

The map a is a bijection.

$$\begin{aligned} a : \{\text{homomorphisms } G \rightarrow \text{Sym}(X)\} &\longrightarrow \{\text{actions } G \curvearrowright X\} \\ \varphi : G \rightarrow \text{Sym}(X) &\longmapsto a(\varphi) : G \times X \rightarrow X : (g, x) \mapsto \varphi(g) \cdot x \end{aligned}$$

Proof. Let us simply construct an inverse. Let $\cdot : G \times X \rightarrow X$ be an action. Define

$$\begin{aligned} \varphi : G &\rightarrow \text{Sym}(X) \\ g &\mapsto \varphi(g) := [g \cdot : X \rightarrow X] \end{aligned}$$

This is a permutation as $(g^{-1}) \cdot : X \rightarrow X$ is the inverse. Indeed,

$$\begin{aligned} \varphi(g^{-1})(\varphi(g)(x)) &= g^{-1} \cdot (g \cdot x) \\ &= (g^{-1}g)x \\ &= x \end{aligned}$$

To see that this rule gives a homomorphism is simple. ■

We have some notation:

- $G^X = \text{im } \phi \leq \text{Sym}(X)$;
- $G_X = \ker \phi \triangleleft G$

It is a special case of the first isomorphism theorem that $G^X \cong G/G_X$.

Example 1.27

Let G be the symmetry group of the unit cube, and let X be the set of the cube's diagonals. Clearly G acts on X . We have hence

$$\phi : G \rightarrow \text{Sym}(X) \cong S_4$$

We can see $\text{im } \phi \cong S_4$, since we can swap any two diagonals – then these transpositions generate the whole symmetric group. Furthermore, $\ker \phi = \{e, s\}$, where s sends every vertex to its opposite. As an immediate consequence, we conclude $|G| = 48$.

Example 1.28

For any group G , we can make G act on itself (as a set) by

$$\begin{aligned} G \times G &\rightarrow G \\ (g, g_1) &\mapsto gg_1 \end{aligned}$$

which gives $\phi : G \rightarrow \text{Sym}(G)$. The kernel $\ker \phi = \{e\}$, and so ϕ is injective – we conclude that G is isomorphic to a subgroup of $\text{Sym}(G)$ by the first isomorphism theorem.

Example 1.29

Let $H \leq G$ and $X = G/H$. Then $G \curvearrowright X$ by

$$(g, g_1H) \rightarrow gg_1H$$

We get a homomorphism $\phi : G \rightarrow \text{Sym}(X)$. Consider the kernel – it is clear that $\ker \phi \subseteq H$. But more is true. If $g \in \ker \phi$, then

$$\begin{aligned} gg_1H &= g_1H \\ \implies g_1^{-1}gg_1 &\in H \\ \implies \ker \phi &\subseteq \bigcap_{g_1 \in G} g_1Hg_1^{-1} \end{aligned}$$

that is, g is contained in every conjugate of H . Furthermore, this containment is reversible: suppose $g \in \bigcap_{g_1 \in G} g_1Hg_1^{-1}$. Then $gg_1H = g_1H$, so $g \in \ker \phi$. We conclude

$$\ker \phi = \bigcap_{g_1 \in G} g_1Hg_1^{-1}$$

If H were normal, we would just have H on the right-hand side – but when H is not normal, we have an interesting way to make a smaller normal subgroup, $\ker \phi$, out of H .

Theorem 1.30

Let G be finite and let $H \leq G$ be of index n . Then, there exists a normal $K \triangleleft G$ with $K \subseteq H$ such that G/K is isomorphic to a subgroup of S_n . Thus $|G/K|$ divides $n!$, and $|G/K| \geq n$.

Proof. Consider $\phi : G \rightarrow \text{Sym}(G/H)$ from the example. The kernel, call it K , is normal. We have shown it is contained in H already – applying the first isomorphism theorem, we have

$$G/K \cong \text{im } \phi \leq \text{Sym}(G/H) \cong S_n$$

Lagrange gives $|G/K|$ divides $n!$, and $K \leq H$, so $|G/K| \geq |G/H| = n$. ■

Corollary 1.31

Let G be a non-abelian simple group. Let $H \leq G$ be a proper subgroup with index $n > 1$. Then G is isomorphic to a subgroup of A_n with $n \geq 5$.

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Claim 3. Any non-trivial $H \triangleleft A_n$ contains a 3-cycle.

- If H contains an element of the form $\sigma = (1\ 2\ \dots\ \gamma)\tau$ with τ disjoint from $(1\ 2\ \dots\ \gamma)$ and $\gamma \geq 4$, then normality of H gives that $\delta^{-1}\sigma\delta \in H$, so furthermore $\sigma^{-1}\delta^{-1}\sigma\delta \in H$. Since τ misses 1, 2, 3, it commutes with δ , and with $(1\ 2\ \dots\ \gamma)$ by assumption.

Then, expanding $\sigma^{-1}\delta^{-1}\sigma\delta$ out, we get $(2\ 3\ \gamma)$, so we have our 3-cycle.

- If H contains an element of the form $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$ in a disjoint cycle decomposition, take $\delta = (1\ 2\ 4)$ and calculate $\sigma^{-1}\delta^{-1}\sigma\delta$ – this is a 5-cycle, bringing us into case 1.
- If H contains $\sigma = (1\ 2\ 3)\tau$ where τ is a product of disjoint transpositions (if τ contained anything longer, we could relabel and apply cases 1 or 2). Then σ^2 is a 3-cycle.
- If H contains $\sigma = (1\ 2)(3\ 4)\tau$ where τ is a product of disjoint transpositions, let $\delta = (1\ 2\ 3)$ and consider $\mu = \sigma^{-1}\delta^{-1}\sigma\delta = (1\ 4)(2\ 3)$. Then let $\nu = (1\ 5\ 2)\mu(1\ 2\ 5) = (1\ 3)(4\ 5)$ – since $\mu\nu = (1\ 2\ 3\ 4\ 5)$, we are in case 1.

This covers all cases up to relabelling.

§1.4 Finite p -groups

Let p be a prime. Groups of order p^n for some n are called p -groups.

Theorem 1.32

Let G be a p -group. Then the centre of G is non-trivial.

Proof. Let G act on itself by conjugation. The centre $Z(G)$ is the set of all size one orbits. Since the orbits partition G , we have

$$|G| = p^n = |Z(G)| + \sum \text{sizes of conjugacy classes of size } > 1$$

We know conjugacy classes have sizes dividing p^n . Hence, if a conjugacy class has size larger than 1, it is a multiple of p . Then $Z(G)$ is a difference of multiples of p and the result follows ■

Theorem 1.33

A group of size p^2 is abelian.

Lemma 1.34

If $G/Z(G)$ is cyclic, then G is abelian.

Proof of lemma. We have a coset $xZ(G)$ in $G/Z(G)$ that generates $G/Z(G)$, so every coset has the form $x^m Z(G)$. This means every G can be written as $g = x^m z$ for some m and $z \in Z(G)$. From this the result immediately follows. ■

Proof of theorem. By [theorem 1.32](#), $Z(G)$ is either G or has size p . In the former case we are done – in the latter case, $G/Z(G) \cong C_p$, which is cyclic, so, by [lemma 1.34](#), G is abelian. ■

Theorem 1.35

Let G be a p -group of size p^n . Then G has a subgroup of size k for all $0 \leq k \leq n$.

Proof. We induct on n . *Base case.* $n = 1$, the result is clear. *Inductive step.* Suppose $n > 1$. If $k = 0$, take $\{e\}$. Otherwise, note $Z(G)$ is non-trivial. Let $x \in Z(G)$ with $x \neq e$, and its order is a power of p – if we replace x with some power of x , say x' , we can produce an element of order p . Then $\langle x' \rangle$ has order p and is normal.

Now consider $G/\langle x' \rangle$. This is a group of size p^{n-1} , and so it contains a subgroup, say L , of size p^{k-1} by induction. By the correspondence of subgroups, we find $K \leq G$ containing $\langle x' \rangle$ such that $K/\langle x' \rangle \cong L$. But then $|K| = p^k$, as required. ■

§1.5 Finite abelian groups**Theorem 1.36 (Classification of finite abelian groups)**

Let G be a finite abelian group. There exist positive integers $d_1, \dots, d_r$ such that

$$G \cong C_{d_1} \times \cdots \times C_{d_r}$$

Furthermore, we can choose the d_i such that $d_{i+1} \mid d_i$, in which case the expression is unique.

Example 1.37

Consider, for example, abelian groups of order 8 – they are exactly C_8 , $C_4 \times C_2$, and $C_2 \times C_2 \times C_2$.

Lemma 1.38 (Chinese remainder theorem)

If n and m are coprime, then $C_n \times C_m \cong C_{nm}$.

Proof. It suffices to find an order mn element of $C_n \times C_m$. If $g \in C_n$ has order n and $h \in C_m$ has order m , then consider $(g, h) \in C_n \times C_m$. Suppose it has order k , then $g^k = h^k = e$, so that $n \mid k$ and $m \mid k$. But then $nm \mid k$, and $k \mid nm$ by Lagrange, so $k = mn$ as required. ■

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Corollary 1.39

Let G be finite and abelian, then we have

$$G \cong C_{l_1} \times \cdots \times C_{l_n}$$

where each l_i is a prime power.

Proof. The result follows from [theorem 1.36](#) and [lemma 1.38](#). ■

§1.6 Sylow theorems

Let G be a finite group of size $p^a m$ where $p \nmid m$ and p is some prime. A *Sylow subgroup* of G is a subgroup of order p^a .

Theorem 1.40 (Sylow theorems)

Let G be a finite group and p a prime.

1. The set $\text{Syl}_p(G) = \{P \leq G : P \text{ is Sylow-}p\}$ is non-empty;
2. any $H, H' \in \text{Syl}_p(G)$ are conjugate;
3. if $n_p = |\text{Syl}_p(G)|$, then $n_p \equiv 1 \pmod{p}$, and $n_p \mid |G|$ (so $n_p \mid m$).

Corollary 1.41

$\text{Syl}_p(G) = \{P\}$ if and only if P is normal.

Proof. For any $g \in G$, gPg^{-1} is isomorphic as a group to P , so it is also a Sylow p -subgroup. ■

Corollary 1.42

Let G be non-abelian and simple and suppose $p \mid |G|$. Then $|G|$ divides $\frac{n_p!}{2}$ and $n_p \geq 5$.

Proof. The conjugation action of G on $\text{Syl}_p(G)$ gives a homomorphism

$$\phi : G \rightarrow \text{Sym}(\text{Syl}_p(G)) \cong S_{n_p}$$

By simplicity of G , $\ker \phi = \{e\}$ or G . If $\ker \phi = G$, then $gPg^{-1} = P$ for all $g \in G$, so P is normal. Since $P \neq \{e\}$, $P = G$. But $Z(P) \triangleleft P$ non-trivially, which is a contradiction. Hence ϕ is injective, so G isomorphic to a subgroup of S_{n_p} . Consider now

$$G \xrightarrow{\phi} S_{n_p} \xrightarrow{\text{sign}} \{\pm 1\}$$

If the composition is surjective, then $\ker(\text{sign} \circ \phi)$ is an index 2 subgroup of G . But G is simple, so this is impossible, (since $G \neq C_2$). In other words, $\text{im } \phi \subseteq A_{n_p}$, and the result follows by Lagrange. Finally, all non-abelian subgroups of A_2 , A_3 and A_4 are not simple. ■

Example

Suppose G has size $11 \times 12 = 132$. If G is simple, then none of the n_p 's are equal to 1. Furthermore, there are 12 Sylow-11 subgroups, since $n_1 1 \equiv 1 \pmod{11}$ and $n_1 1 \mid 12$.

Similarly, $n_3 \equiv 1 \pmod{3}$, and $n_3 \mid 44$, so $n_3 = 4$ or 22 – by [corollary 1.42](#), $n_3 = 22$. Any two Sylow 11-subgroups intersect only at $\{e\}$, so we already have 120 elements. We also have at least 44 elements of order 3, which is impossible, so no such simple G exists.

Proof of Sylow part 1. We will try and produce a Sylow p -subgroup as a stabiliser of an element in an appropriate action.

Let $|G| = n = p^a m$, of course $p \nmid m$. Let $\Omega = \{X \subseteq G : |X| = p^a\}$. Let G act on Ω by $g \cdot \{g_1, \dots, g_{p^a}\} = \{gg_1, \dots, gg_{p^a}\}$.

Claim 1. We have $|\Omega| = \binom{n}{p^a} = m \not\equiv 0 \pmod{p}$.

Proof. Expand $(1+x)^n \pmod{p}$.

Claim 2. Suppose $U \in \Omega$ and $H \leq G$ stabilises U , then $|H| \mid |U|$.

Proof. We have $hU = U$ for all $h \in H$. In other words, for all $u \in U$, the coset Hu is contained in U . Furthermore, every $u \in U$ lies in some coset of H , and the cosets are disjoint, so the cosets partition U . We get $|H| \mid |U|$.

We now know $|\Omega| \not\equiv 0 \pmod{p}$, and $|\Omega| = |O_1| + \dots + |O_r|$ where the O_i are the orbits of the action. Since $p \nmid |\Omega|$, there exists an orbit, say Θ , whose size is coprime to p .

Let $T \in \Theta$. By the orbit-stabiliser theorem, $|G| = |\Theta| \cdot |\text{Stab}(T)|$, so $p^a m = |\Theta| |\text{Stab}(T)|$. By claim 2, $|\text{Stab}(T)|$ divides p^a , but there are no factors of p in Θ , so we are done. ■

Lecture n

§2 Rings

Lecture $n + 1$

§2.1 Basic definitions

Let R be a ring. We have

$$\begin{aligned} 0 + 0 &= 0 \\ \implies r \cdot 0 + r \cdot 0 &= r \cdot 0 \\ \implies r \cdot 0 &= 0 \end{aligned}$$

Furthermore, if $1 = 0$, we have $r = 1 \cdot r = 0 \cdot r = 0$, so we have the *zero ring*.

Definition 2.1 (Ring direct product). Let R, S be rings. The *direct product* $R \times S$ is a ring with operations and identities defined pointwise.

Definition 2.2 (Polynomials). Let R be a ring. A *polynomial* in X with coefficients in R is an expression of the form

$$f(X) = a_0 + a_1X + \cdots + a_nX^n$$

where each $a_i \in R$, and the X^i are formal symbols. As convention, we will identify $f(X)$ with $f(X) + 0 \cdot X^{n+1}$ – then, the largest n with $a_n \neq 0$ is the *degree* of $f(x)$ (by convention, we say the degree of 0 is $-\infty$). Finally, a polynomial is *monic* if it has degree n and $a_n = 1$.

Definition 2.3 (Polynomial rings). Let R be a ring. The *polynomial ring* $R[X]$ is the set of all polynomials with coefficients in R , with obvious operations and identities.

Remark. Note that, for every $f \in R[X]$, we get a function $R \rightarrow R$ which takes r to $f(r)$. But this is not the same as the polynomial itself.

Definition 2.4 (Rings of formal power series). The ring of *formal power series* over R in the variable X is

$$R[[X]] = \left\{ \sum_{i=0}^{\infty} r_i X^i : r_i \in R \right\}$$

with obvious operations and identities.

Example 2.5

Consider the element $1 - x \in R[X]$. Is it a unit?

Suppose we have $(1 - x)g(x) = 1$, where $g(x) = a_0 + \cdots + a_n x^n$, with $a_n \neq 0$. Then

$$(1 - x)g(x) = a_0 + (a_1 - a_0)x + \cdots + (a_n - a_{n-1})x^n - a_n x^{n+1}$$

and then, by comparing the x^{n+1} coefficient, we can see this is not 1.

On the other hand, if we view $1 - x \in R[[X]]$, this has an inverse:

$$(1 - x)(1 + x + x^2 + \cdots) = 1$$

In fact, every power series has an inverse as long as r_0 is invertible.

Definition 2.6 (Rings of Laurent series). Write $R[X, X^{-1}]$ for the ring of *Laurent series* with coefficients in R , that is

$$f = \sum_{i \in \mathbb{Z}} a_i x^i$$

with $a_i \in R$ and all but finitely many $a_i = 0$.

§2.2 Homomorphisms, ideals and quotients

Definition 2.7 (Ring homomorphisms). Let R and S be rings. A function $f : R \rightarrow S$ is a *ring homomorphism* if

- $f(r_1 + r_2) = f(r_1) + f(r_2)$;
- $\implies f(0) = 0$;
- $f(r_1 r_2) = f(r_1) f(r_2)$;
- $\not\implies f(1) = 1$.

The kernel of f is $\ker f = \{r \in R : f(r) = 0\}$, and the image of f is $\operatorname{im} f = \{s \in S : s = f(r) \text{ for some } r \in R\}$.

Definition 2.8 (Ring isomorphisms). If $f : R \rightarrow S$ is a bijective ring homomorphism, then it is called an isomorphism. Furthermore, its inverse is also a ring homomorphism.

Definition 2.9. A subset $I \subseteq R$ is called an *ideal*, written $I \triangleleft R$, if

- I is a subgroup additively;
- if $a \in I$ and $b \in R$, then $ab \in I$.

Usually ideals are not subrings, since they can only contain 1 if $I = R$.

Lemma 2.10

If $\phi : R \rightarrow S$ is a homomorphism, then $\ker \phi \triangleleft R$.

Proof. Since ϕ is a group homomorphism, we know $\ker \phi$ is a subgroup.

Furthermore, if $a \in \ker \phi$ and $b \in R$, then $\phi(ab) = \phi(a)\phi(b) = 0 \cdot \phi(b) = 0$. ■

Remark. If $a \in R$ is a unit, and $I \triangleleft R$ is an ideal, then $u \in I \implies I = R$ – since $1 = uv$, $1 \in I$, and so $I = R$.

Example 2.11

Observe that $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ for any n . In fact, every ideal of $\mathbb{Z}$ has this form. Indeed, if $I \neq \{0\}$, let $n \in \mathbb{Z}$ be the minimal positive element of I . By the Euclidean algorithm, we can write $m \in I$ as $m = qn + r$, with $0 \leq r < n$, and $qn \in I$, so $r \in I$ – this contradicts minimality of n .

Definition 2.12. Let $A \subseteq R$ be a subset. The *ideal generated by A* is

$$\langle A \rangle = \left\{ \sum_{a \in A} r_a a : r_a \in R, r_a = 0 \text{ for all but finitely many } a \right\}$$

Definition 2.13. An ideal is called *principal* if there exists $r \in R$ such that $I = \langle r \rangle$.

Example 2.14

Consider the subset of $R[x]$ given by

$$\{f(x) \in R[x] : f(0) = 0\}$$

which is generated by $x \in R[x]$ – hence, this is a principal ideal.

Definition 2.15 (Quotients). Let $I \triangleleft R$ be an ideal. The *quotient ring* R/I is the set of cosets of I in R with 0 and 1 given by $0_R + I$ and $1_R + I$, with the rules

$$\begin{aligned} (r_1 + I) + (r_2 + I) &= (r_1 + r_2) + I \\ (r_1 + I) \cdot (r_2 + I) &= r_1 r_2 + I \end{aligned}$$

Proposition 2.16

The quotient ring is a ring. Furthermore, the function

$$\begin{aligned} f : R &\rightarrow R/I \\ r &\mapsto r + I \end{aligned}$$

is a ring homomorphism.

Proof. This is obviously a group homomorphism. Furthermore, multiplication is well-defined: suppose $r_1 + I = r'_1 + I$, $r_2 + I = r'_2 + I$. We then have $r_i - r'_i \in I$, say $r_i - r'_i = a_i$. We then have

$$r'_1 r'_2 = (r_1 + a_1)(r_2 + a_2) = r_1 r_2 + \underbrace{r_1 a_2 + r_2 a_1 + a_1 a_2}_{\in I}$$

We conclude that the coset in I of $r'_1 r'_2$ is the same as that of $r_1 r_2$.

To check the other ring properties is straightforward. ■

Example 2.17 • We have $n\mathbb{Z} \triangleleft \mathbb{Z}$. The quotient ring $\mathbb{Z}/n\mathbb{Z}$ consists of cosets of $n\mathbb{Z}$, i.e. $0 + n\mathbb{Z}$, $1 + n\mathbb{Z}$, $\dots$, $(n-1) + n\mathbb{Z}$. The ring structure is modular arithmetic.

- Take $\langle x \rangle \triangleleft \mathbb{C}[x]$. The elements of the quotient $\mathbb{C}[x]/\langle x \rangle$ can be represented by elements of the form $(a_0 + a_1 x + \dots + a_n x^n) + \langle x \rangle$ – but all the terms $a_1 x, \dots, a_n x^n$ lie in $\langle x \rangle$, so every element is uniquely represented by a_0 . Hence $\mathbb{C}[X]/\langle x \rangle \cong \mathbb{C}$.

Proposition 2.18 (Euclidean algorithm for polynomial rings over fields)

Let K be a field, and take $f, g \in K[x]$. There exist polynomials $r, q \in K[x]$ such that

$$f = gq + r$$

and $\deg r < \deg g$.

Proof. Let $n = \deg f$ and $m = \deg g$, so

$$f = \sum_{i=0}^n a_i x^i \quad g = \sum_{i=0}^m b_i x^i$$

with $a_i, b_i \in K$, $a_n \neq 0$. If $n < m$, set $q = 0$ and $r = f$. On the other hand, if $n \geq m$, we proceed by induction on n . In particular, let $f_1 = f - a_n b_m^{-1} x^{n-m}$ (note we are using the fact that K is a field). Now, $\deg f_1 < n$. If $n = m$, we can write

$$f = (a_n b_m^{-1} x^{n-m})g + f_1$$

Otherwise, if $n > m$ we can write by induction

$$f_1 = gq_1 + r_1$$

with $\deg r_1 < m$. Then

$$f = (a_n b_m^{-1} x^{n-m} + q)g + r_1$$

■

Corollary

All ideals in $K[x]$ for K a field are principal.

Proof. We have proven that Euclidean algorithm $\implies$ all ideals are principal, since we did it for $\mathbb{Z}$. The result follows. ■

Theorem 2.19 (First isomorphism theorem for rings)

Let $\phi : R \rightarrow S$ be a ring homomorphism. Then, the map

$$\begin{aligned} f : R / \ker \phi &\rightarrow \text{im } \phi \\ r + \ker \phi &\mapsto \phi(r) \end{aligned}$$

is a well-defined ring isomorphism.

Proof. Well-definedness, bijectivity and additivity all follow from the group isomorphism theorem. It remains to check multiplicativity. Observe

$$f((r + \ker \phi)(t + \ker \phi)) = f(rt + \ker \phi)$$

$$\begin{aligned}
 &= \phi(rt) \\
 &= \phi(r)\phi(t) \\
 &= f(r + \ker \phi)f(t + \ker \phi)
 \end{aligned}$$

■

Example 2.20

Consider the homomorphism

$$\begin{aligned}
 \phi : \mathbb{R}[x] &\rightarrow \mathbb{C} \\
 f(x) &\mapsto f(i)
 \end{aligned}$$

The map is clearly surjective, and the kernel is exactly those polynomials f satisfying $f(i) = 0$. However, since coefficients of f are real, $-i$ is also a root. In conclusion, $x^2 + 1 \mid f$, and we conclude $\ker \phi = \langle x^2 + 1 \rangle$. Hence, by the first isomorphism theorem, $\mathbb{C} \cong \mathbb{R}[x] / \langle x^2 + 1 \rangle$.

Proposition 2.21 (Second isomorphism theorem for rings)

Let $R \leq S$ and $J \triangleleft S$. Then $J \cap R$ is an ideal in R , and

$$(R + J)/J = \{r + J : r \in R\} \leq S/J$$

is a subring. Furthermore,

$$R/(R \cap J) \cong (R + J)/J$$

Proof. Define a function

$$\begin{aligned}
 \phi : R &\rightarrow S/J \\
 r &\mapsto r + J
 \end{aligned}$$

so $\ker \phi = \{r \in R : r + J = 0\} = R \cap J$, so this is an ideal by the first isomorphism theorem. Also, the image is $\text{im } \phi = \{r + J : r \in R\} = (R + J)/J$. The result follows from the first isomorphism theorem. ■